
```
clc;
clear;
close all;
```

Parameters

```
N = 100;           % Number of bits
SNR = 10;          % Signal-to-Noise Ratio (dB)
```

Generate Random Bits

```
bits = randi([0 1], N, 1);
```

BPSK Modulation

0 -> -1 1 -> +1

```
bpsk = 2*bits - 1;
```

Add AWGN Noise

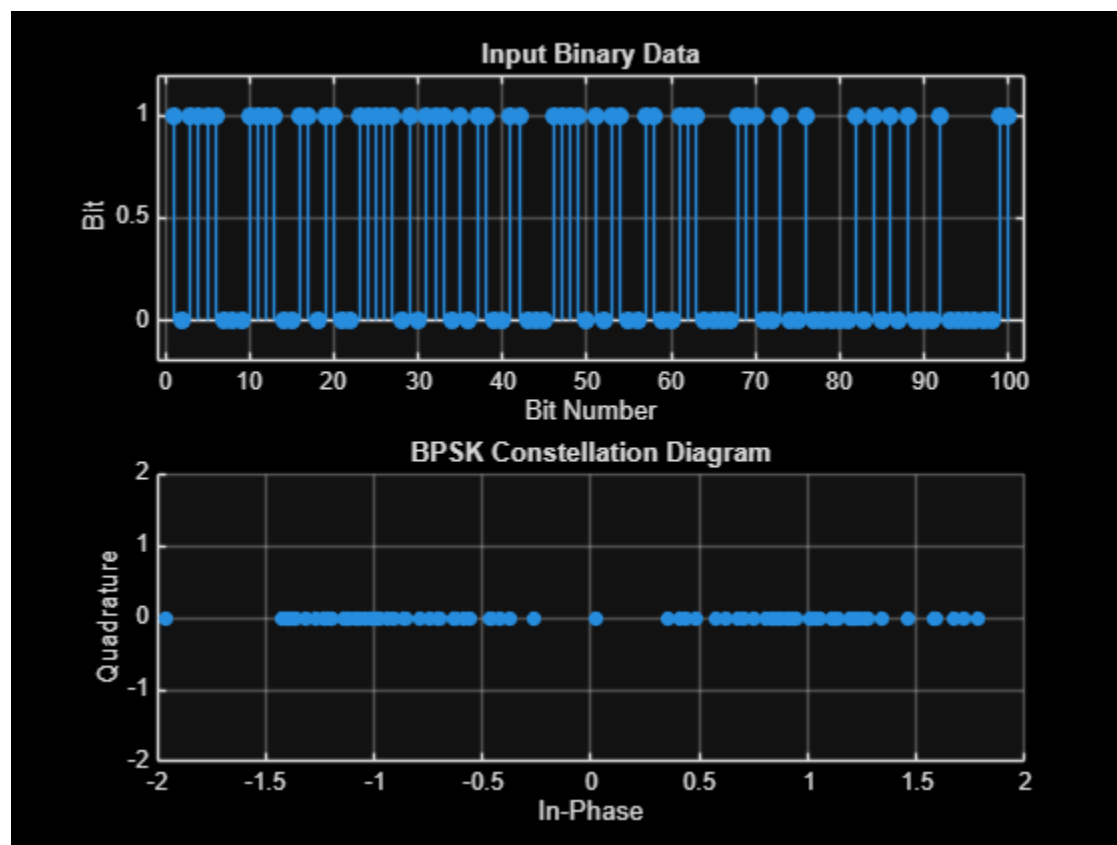
```
rx = awgn(bpsk, SNR, 'measured');
```

Plot Results

```
figure('Name', 'BPSK Modulation', 'NumberTitle', 'off');
```

```
% Input Binary Data
subplot(2,1,1);
stem(bits, 'filled');
title('Input Binary Data');
xlabel('Bit Number');
ylabel('Bit');
grid on;
ylim([-0.2 1.2]);
```

```
% BPSK Constellation Diagram
subplot(2,1,2);
scatter(real(rx), imag(rx), 30, 'filled');
title('BPSK Constellation Diagram');
xlabel('In-Phase');
ylabel('Quadrature');
grid on;
axis([-2 2 -2 2]);
```



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